Nitesh Kumar

Fourth Year Undergraduate | Department of Electrical Engineering
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Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2023-Present	B.Tech-Electrical Engineering	Indian Institute of Technology Kanpur	7.0/10
2022	Class XII (RBSE)	M.D. Sr. Sec. School, Chirawa	93.20%
2020	Class X (RBSE)	M.D. Sr. Sec. School, Chirawa	90.17%

Work Experience

EleWin | **Backend Developer** | `Elewin.in` *Apr'26 – Jul'26*

- Built **REST APIs** for the **recruiter platform** using **Node.js**, **Express.js**, and **MongoDB** to manage job roles
- Developed **candidate search, filtering, and profile APIs** to support recruiter workflows for reviewing and shortlisting
- Implemented reusable **JWT authentication, validation, and RBAC middleware** for protected recruiter APIs
- Designed APIs for **job and candidate status management**, supporting workflows such as review, shortlisting, and updates
- Developed **recruiter dashboard APIs** to retrieve candidate and job data using MongoDB queries and error handling

Tech Stack: Node.js, Express.js, MongoDB, JWT

Key Projects

Digitizing Entry With Flutter | **Project Mentor, ACA** *(May'25 - Jul'25)*

- Developed an **Admin App** using **Flutter** and **Firebase** to manage ticket pricing, time slots, and booking-related operations
- Implemented **Firebase Authentication** for secure admin login and access to protected management features
- Built **ticket management workflows** for creating, updating, and tracking ticket details through the admin dashboard
- Integrated **QR-based ticket verification** to allow administrators to validate customer tickets during entry
- Implemented **Cloud Firestore** integration for storing and retrieving ticket, pricing, and time-slot data across the application

Tech Stack: Flutter, Firebase, Firebase Auth, Cloud Firestore, QR Flutter

AI 101 | **Electrical Engineering Association, IIT Kanpur** *(Dec'24- Feb'25)*

- Fine-tuned a pre-trained **BERT** model on customer reviews to perform binary sentiment classification for real-world data
- Prepared text data through **cleaning, tokenization,** and train-validation splitting to create suitable inputs for training
- Trained the model on task-specific reviews and evaluated predictions using **Accuracy** and **F1-score** metrics for assessment
- Used the fine-tuned model to classify new customer reviews as **Positive** or **Negative** sentiments during inference in practice

Tech Stack: Python, Pandas, NumPy, Scikit-learn, PyTorch, Hugging Face Transformers

Employee Attrition Predictor | **Course Project - EE708** | **Prof. Rajesh M. Hegde** *(May'25 - July'25)*

- Built an employee attrition prediction model utilizing key feature extraction on the **IBM HR Analytics** dataset using **Logistic Regression** and **Random Forest**, achieving **74% accuracy** for predicting employee turnover risk
- Performed **data cleaning, categorical encoding,** and **feature scaling**, with **5-fold cross-validation** for model evaluation
- Compared models using **Precision, Recall, F1-score,** and **ROC-AUC** for overall performance evaluation

Tech Stack: Python, Pandas, NumPy, Scikit-learn, Logistic Regression, Random Forest, Matplotlib, Seaborn

In-Memory Key-Value Store | **Self Project** | *(May'26 - Jul'26)*

- Developed an in-memory key-value store with **unordered_map** and **unordered_set**, enabling **O(1)** average-time operations
- Structured into **Command Parser, Storage Engine, Expiry Manager,** and **Database** modules following **OOP** principles
- Added automatic key expiration with **C++ chrono** and file persistence to recover database state after application restarts

Tech Stack: C++, STL, Object-Oriented Programming (OOP), `unordered_map`, `chrono`, File I/O

Stock Price Forecasting and Analysis | **Self Project** | *(May'24- Jul'24)*

- Applied **feature engineering** to financial data using moving averages, momentum, and lagged returns for forecasting
- Trained **Random Forest** and **LSTM** models, achieving **90% accuracy**, visualized metrics using **Matplotlib** and **Plotly**

Technical Skills

Programming Languages	C++, C, Python, JavaScript, TypeScript, SQL, Dart, LaTeX
Backend & Databases	Node.js, Express.js, MongoDB, Redis, BullMQ, Firebase, Cloud Firestore, REST APIs
AI/ML & Computer Vision	Scikit-learn, TensorFlow, Pandas, NumPy, Matplotlib, Seaborn, OpenCV, Statsmodels
AI & APIs	Google Gemini API, OpenAI API, DALL-E 3, Flux API, Kling API, PixVerse API
Frontend & Mobile	React, Next.js, Flutter, Chart.js, Mermaid
Developer Tools	Git, GitHub, Docker, VS Code, Firebase Storage, Firebase Auth, ObjectBox

Relevant Courses

Fundamentals of Computing	Probability and Statistics	Data Structures and Algorithms
Software Project Management	Linear Algebra	Partial Differential Equations
Complex Variables	Ordinary Differential Equations	Signal Systems and Networks

Extra-Curriculars

Sports	• Secured **2nd prize in Hockey** at Inferno 24, IIT Kanpur's annual inter-pool sports championship event • Participated in **Cricket** at Inferno 24, IIT Kanpur's annual inter-pool sports championship event
Cultural	• Contributed to **Galaxy'24**, participating in events and representing the hall in the **Galaxy Parade**